improvements targeted at digital painters and photographers, metadata editing and much more, including an updated user interface and initial HiDPI support.

One thing immediately noticeable about GIMP 2.10 is the new dark theme and symbolic icons enabled by default. This is meant to somewhat dim the environment and shift the focus towards content.

There are now four user interface themes available in the GIMP—dark (default), grey, light, and system. Icons are now separate from themes, and both colour and symbolic icons are present, so the GIMP can be configured to have the system theme with coloured icons if the old look is preferred.

Further, icons are available in four sizes now, so that GIMP looks better on HiDPI displays. The GIMP will do its best to detect which size to use, but one can manually override that selection as follows: *Edit > Preferences > Interface > Icon Themes.*

The ultimate goal for v2.10 was completing the port to the GEGL image processing library, which had started with v2.6 (when the optional use of GEGL for colour tools and an experimental GEGL tool were introduced) and continued with v2.8 (when GEGL based projection of layers was added).

Now, the GIMP uses GEGL for all tile management and builds an acyclic graph for every project. This is a prerequisite for adding the non-destructive editing planned for v3.2.

There are many benefits that accrue from using GEGL, and some of them can already be enjoyed in GIMP 2.10.

High bit depth support allows the processing of images with up to 32-bit per colour channel precision and of open/export PSD, TIFF, PNG, EXR and RGBE files in their native fidelity. Additionally, FITS images can be opened with up to 64-bit per channel precision.

Multi-threading allows making use of multiple cores for processing. Not all features in the GIMP make use of that, though. A point of interest is that multi-threading happens through GEGL processing, but also in core GIMP itself

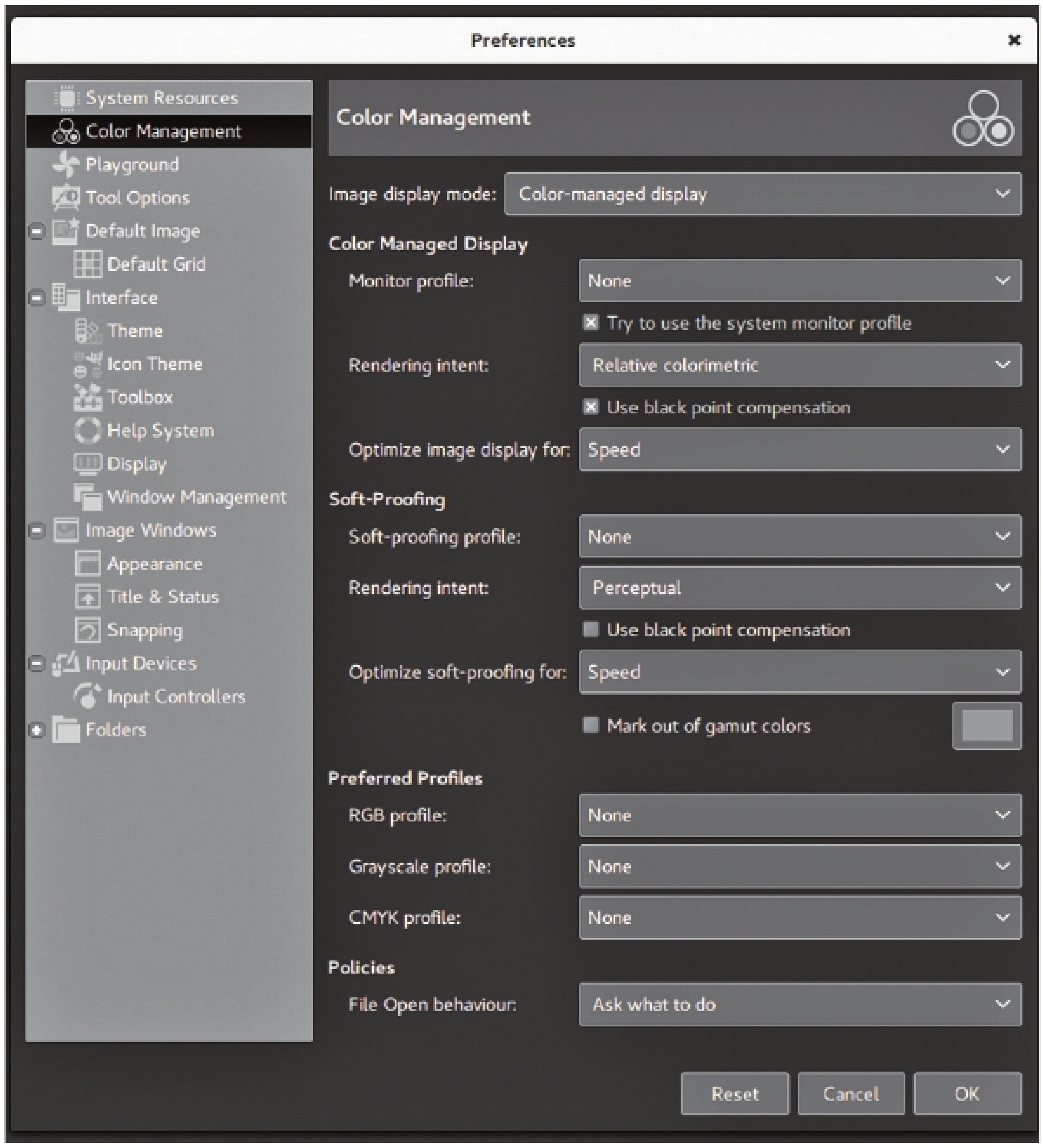

Figure 5: Colour management

—for instance, to separate painting from display code.

GPU-side processing is still optional but available for systems with stable OpenCL drivers. One can find configuration options for multi-threading and hardware acceleration by following *Edit > Preferences > System Resources.*

Another benefit of using GEGL is being able to work on images in a linear RGB colour space as opposed to a gamma-corrected (perceptual) RGB colour space.

Here is what it boils down to—you now have both linear and perceptual versions of most blending modes. There is now a linear version of the *colour invert* command. You can freely switch between the two at any time via the *Image > Precision* sub-menu. You can choose which mode is displayed in the Histogram docker. You can apply filters for levels and curves in either perceptual or linear mode. When higher than 8-bit per channel precision is used, all channel data is linear. You can choose whether the gradient tool should work in perceptual RGB, linear RGB or CIE LAB colour space.

Colour management is now a core feature of the GIMP rather than a plugin. This has made it possible to introduce colour management to all custom widgets—image previews, colour and pattern previews, etc.

The GIMP now uses LittleCMS v2, which allows it to use ICC v4 colour profiles. It also partially relies on the *babl* library for handling colour transforms, since *babl* is up to ten times faster than LCMS2. Eventually, *babl* could replace LittleCMS in the GIMP. The latter now ships with two groups of blending modes—legacy (perceptual, mostly to